

SWARNALI MOLICK

DATA SCIENTIST | MACHINE LEARNING & APPLIED AI | MSc AI & DATA ENGINEERING

Venice, Italy | Available remotely | swarnalimollick@gmail.com | linkedin.com/in/s-mollick | github.com/itsmeSwarnali

PROFESSIONAL SUMMARY

AI and Data Engineering MSc candidate with hands-on experience building end-to-end data science, machine-learning, NLP and Applied AI systems. Strong in Python, SQL, data analysis, statistical modelling, model evaluation, APIs and deployment. Combines peer-reviewed machine-learning research with practical projects for junior Data Science, Machine Learning and Applied AI opportunities in Europe.

CORE TECHNOLOGIES

Data & Programming	Python, SQL, R, Pandas, NumPy, PySpark, Apache Spark, EDA, feature engineering
Machine Learning	Scikit-learn, XGBoost, Random Forest, classification, forecasting, cross-validation, evaluation
Applied AI	PyTorch, TensorFlow/Keras, Transformers, DistilBERT, Llama, LangChain, RAG, embeddings, ChromaDB
Deployment	FastAPI, Docker, Streamlit, Flask, Git/GitHub, REST APIs, technical documentation

SELECTED PROJECTS

Customer Retention Predictor

- Built a complete telecom churn workflow for 7,043 customers; Logistic Regression achieved 0.845 ROC-AUC and 80.4% accuracy. Deployed preprocessing and prediction through FastAPI and Docker.

Sales Prediction Tool

- Analysed 1M+ Rossmann sales records and compared five regression models. Random Forest achieved R^2 0.868; promotions increased average sales by approximately 39%.

Financial News Analyzer

- Compared fine-tuned DistilBERT with few-shot Llama 3.1 for bullish, bearish and neutral financial text. DistilBERT achieved about 85.8% accuracy and 82.0% Macro-F1.

AI Study Assistant

- Built and deployed a document-grounded RAG assistant using LangChain, ChromaDB, Hugging Face embeddings, Llama/Groq, intent classification and Streamlit Cloud.

Air Quality Predictor

- Built a nonlinear statistical model from pollutant and weather data; the selected Box-Cox GAM explained approximately 93% of variation.

Image Description Generator

- Developed a multi-encoder PyTorch captioning pipeline with ResNet features, spatial attention, LSTM decoding and beam search; best BLEU-4 about 26.9.

PROFESSIONAL EXPERIENCE

Web Developer / Web Programmer - Appstick, Bangladesh

Sep 2022 - Sep 2023

- Developed and maintained responsive, database-backed web applications using Laravel, PHP, MySQL, HTML, CSS and Bootstrap.

Web Designer Intern - The Virtual BD, Bangladesh

Apr 2021 - Feb 2022

- Designed responsive WordPress and Elementor pages and received the Best Web Designer Award for internship performance.

EDUCATION

MSc in Computer Science and Information Technology - AI and Data Engineering

Ca' Foscari University of Venice, Italy | 2024 - 2026 | Italian Government MAECI Scholar

- Coursework includes Machine Learning, Artificial Intelligence, Advanced Data Management, Deep Learning for NLP, Time Series and Optimisation.

BSc in Computer Science and Engineering

Northern University of Business and Technology Khulna | 2018 - 2022

- Undergraduate research included deep ensemble learning for osteosarcoma classification.

RESEARCH & SELECTED PUBLICATIONS

Co-author of 10+ journal and conference publications in healthcare AI, machine learning, data mining, association-rule mining and environmental prediction. Full record available through Google Scholar.

- Adapted deep ensemble learning-based voting classifier for osteosarcoma cancer classification - Diagnostics, 2023.
- A comparative study, prediction and development of chronic kidney disease using machine learning - Human-Centric Intelligent Systems, 2023.
- An unsupervised cluster-based feature grouping model for early diabetes detection - Healthcare Analytics, 2022.
- Efficient prediction of water quality index using machine-learning algorithms - Human-Centric Intelligent Systems, 2021; Best Paper Award.

HONOURS & AWARDS

- Italian Government MAECI Scholarship, 2024-2026.
- Best Paper Award, Human-Centric Intelligent Systems.
- Best Web Designer Award, The Virtual BD, 2021.
- NUBTK Departmental Scholarship for Excellent Results, 2018 and 2019.

LANGUAGES

Bengali (native) | English (IELTS 7.0) | Italian (A1, actively learning) | Hindi (conversational)